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Natural Language Processing Approaches in Industrial Maintenance: A Systematic Literature Review

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Abstract

Industrial maintenance plays a crucial role in manufacturing by significantly reducing machine failure time and minimizing costs, especially in the revolution of Industry 4.0. Consequently, researchers and industrial engineers have continuously focused on this area. Manufacturing companies possess extensive maintenance reports or logs containing valuable textual information, which offers a new avenue for exploring effective industrial maintenance methods. Natural Language Processing (NLP), a subfield of Artificial Intelligence, has demonstrated remarkable potential in analyzing maintenance reports and achieving promising results in various tasks. This paper presents a comprehensive systematic literature review that specifically concentrates on the applications of NLP approaches employed in the field of industrial maintenance. Additionally, this review analyzed the datasets utilized in previous studies and the evaluation measures adopted, which can serve as a valuable resource for other researchers seeking potential solutions in maintenance. Furthermore, the paper discusses the challenges encountered in applying NLP to industrial maintenance and outlines future research directions in this domain. By conducting this systematic literature review, we provide a comprehensive understanding of the current state of NLP applications in industrial maintenance, identify gaps in the existing literature, and guide future research efforts in leveraging NLP techniques for enhanced maintenance practices.

Keywords: Artificial intelligence; AI; Natural language processing; NLP; Industrial maintenance; Systematic literature review.

1. Introduction

Currently, the industry is undergoing a huge revolution and entering a new phase as known as Industry 4.0 [1], [2], which aims to make the industrial systems autonomous [3], [4], [5]. The integration of physical and digital systems in production environments is a central aspect of Industry 4.0 [2]. A large amount of data from various machines/parts

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in industrial systems are required to achieve this integration. These data allow different formats such as numerical data, textual data, graphical data, etc., collected from the entire lifecycle of the production line. With effective processing and analysis, they can provide valuable information to perform a great many significant activities such as decision-making, product design, machine failure detection and others [6], [7].

Maintenance plays a vital role in manufacturing and encompasses routine inspections, equipment repairs, part replacements, and other services [8]. It directly affects production line operations and is closely linked to production efficiency. According to Forbes magazine, manufacturers experience an average of over 15 hours of downtime per week, totalling 800 hours per year. In monetary terms, automotive manufacturers lose more than £18,000 per minute during production line breakdowns. The estimated annual cost of downtime in manufacturing exceeds £400 billion. Effective maintenance measures can significantly reduce unnecessary downtime, resulting in substantial cost savings. Previous literature [9], [10], [2] has classified maintenance strategies into three categories: corrective maintenance (run-to-failure), preventive maintenance, and predictive maintenance:

- **Corrective maintenance/Run-to-failure:** It is the simplest strategy in maintenance, which occurs after equipment breaks down and needs to be repaired or replaced. It could lead to high downtime and numerous defective products may be produced.
- **Preventive maintenance:** It occurs according to the plan with periodic inspection and maintenance execution. The production line still runs when the maintenance activities are performed. It can help predict possible failures and therefore reduce the probability of failures. However, this scheduled maintenance is not necessary when it happens without a fault in the production line, which may increase the extra cost.
- **Predictive maintenance:** It is based on continuous condition monitoring of a machine or a production process, and it happens only when faults occur, or maintenance actions are necessary. Using historical data, it is able to perform early detection of bad health states. In predictive maintenance, a large number of methods are employed like machine learning, statistical techniques and traditional engineering techniques. predictive maintenance has received much attention from researchers because it can significantly improve the efficiency of production lines.

Generally, each manufacturing company has an enormous number of maintenance reports/logs or maintenance work orders that contains maintenance time, equipment or machine, maintenance process, etc., which can help technicians finish a variety of tasks in production lines [11], [12], [13]. The considerable quantity of textual information provides new opportunities to explore industrial maintenance approaches. In recent years, these various types of text can be analysed using natural language processing (NLP) approaches [14], [15], [16], [17]. NLP, a critical component of artificial intelligence, demonstrates significant potential in unlocking insights from textual data. NLP generally consists of two key components: natural language understanding and natural language generation [18]. Its primary goal is to process and analyze text, enabling a range of tasks such as text classification, text summarization, sentiment analysis, etc. In industrial maintenance, NLP offers a promising approach to handle maintenance texts by mining textual data, extracting semantic features, and manipulating information. NLP can facilitate various maintenance tasks, including failure prediction, cause analysis, and decision support. By harnessing NLP techniques, manufacturing companies can potentially increase the overall efficiency of their production lines and avoid unnecessary costs, ultimately enhancing their operational effectiveness.

While several papers have explored the application of NLP methods in maintenance, the selection of appropriate NLP techniques remains an unexplored problem that can significantly impact application performance. Therefore, this paper aims to conduct a systematic literature review that encompasses existing solutions for maintenance based on NLP approaches, along with outlining potential future research directions. Additionally, the selected datasets and evaluation measures employed in these studies will be thoroughly discussed to foster standardization within the field. For this purpose, this paper attempts to address the following research questions:

- RQ1: What are NLP approaches applied in industrial maintenance?
- RQ2: How does NLP approaches solve maintenance problems?
- RQ3: What are the current challenges of NLP approaches in industrial maintenance combining the analysis of employed datasets and evaluation measures?
- RQ4: What future research needs to be conducted to address existing challenges of NLP approaches in industrial maintenance?

The remainder of this article is structured as follows: Section 2 describes the literature selection process and statistical analysis conducted on the selected papers. Section 3 provides an overview of NLP approaches applied in the context of industrial maintenance. The last part, Section 4, gives a description of datasets and evaluation measures used in papers.

2. Systematic literature review

2.1. Literature selection

The systematic literature selection process is illustrated in Figure 1. Research papers are extracted from Scopus and ScienceDirect databases as these two databases cover a broad range of peer-reviewed articles in the academic domain. The search string depends on a careful reading of relevant papers, and Boolean operators are used to enhance the results of literature search. The search string is written as follows:

TITLE-ABS-KEY ((“maintenance” OR “health management” OR “fault diagnosis”) AND (“natural language processing” OR “NLP” OR “text mining”))

The time span for the literature review is set from 2018 to 2022. After the first-round search, 549 items from Scopus and 116 items from ScienceDirect were retrieved (accessed on Oct 30, 2022). Then, the second-round search was conducted, focusing only on journals and conference papers in English. Additionally, duplicates were removed, and papers not relevant to industrial maintenance were excluded in this round. In the second round, 48 papers were extracted. Finally, an in-depth review was performed by reading the contents thoroughly, and papers without natural language processing techniques were excluded. A total of 27 items were selected for further analysis. The systematic literature selection process is depicted in Figure 1.

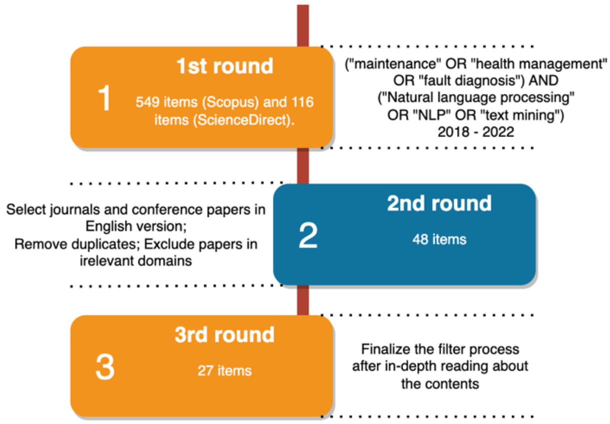

Fig. 1. The systematic literature selection process.

2.2. Statistical analysis of selected papers

Figure 2 presents the number of papers published each year from 2018 to 2022 according to the initial search results. From the figure, it is evident that the number of papers on the application of natural language processing in manufacturing maintenance shows an increasing trend year by year, indicating a common concern among researchers recently. The most significant increase in the number of research papers occurred in 2020-2021, which may be related to the growing volume of text data processed by manufacturing companies and the recent significant advancements in NLP technology. Furthermore, the selected papers, which focus on the application of NLP in industrial maintenance, have been published in a wide range of journals and conferences with an engineering orientation, as shown in Figure 3. Among the 27 papers, 18 were published in journals and 9 in conferences. These research papers are spread across

16 different journals. Notably, *Aerospace* and *IEEE Access* are the top journals with 2 publications each, while other journals have 1 publication each. Most of these journals belong to the manufacturing and engineering domain, although some, like *IEEE Transactions on Industrial Informatics* and *Enterprise Information Systems*, emphasize the application of informatics. Additionally, 9 conference papers are derived from 8 different conferences. *IEEE International Conference on High Voltage Engineering and Application* is a conference with two publications while other conferences only have one publication. These conference papers span a wide range of domains, including control, engineering applications, computer systems, and others, highlighting the evolving intersection of manufacturing and emerging information technology.

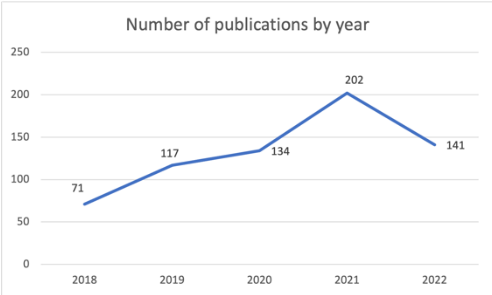

Fig. 2. Number of publications by year from 2018 to 2022.

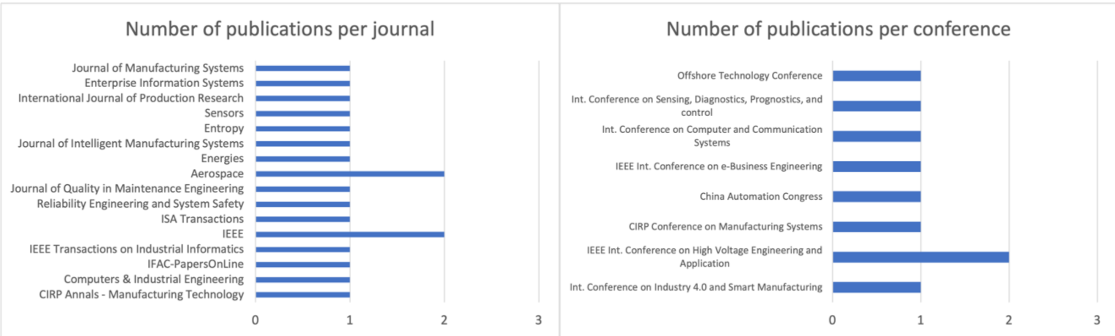

Fig. 3. Number of publications per journal or per conference.

3. NLP approaches in industrial maintenance

From selected papers, it is known that various NLP approaches have been applied in the field of industrial maintenance. Figure 4 summarizes the different NLP approaches employed in these papers and their applications in industrial maintenance, and Table 1 provides a comprehensive summary of the selected papers, organizing them according to various criteria, including maintenance categories, data size, data source, specific purpose, and evaluation measures. From Table 1, it is known that some research studies have focused on analyzing the entire production line, while others have a more specific focus on individual pieces of equipment or machinery. Additionally, predictive maintenance emerges as a particularly promising approach among the three maintenance strategies considered. Since predictive maintenance enables continuous production in the production line based on condition monitoring, it not only saves considerable costs but also contributes to increased productivity and efficiency in industrial processes.

In general, NLP approaches can be grouped into four categories: rule-based approaches, word embedding approaches, machine learning/deep learning approaches, and knowledge graph approaches. This section will provide a detailed introduction to the various NLP techniques and applications used in the selected publications.

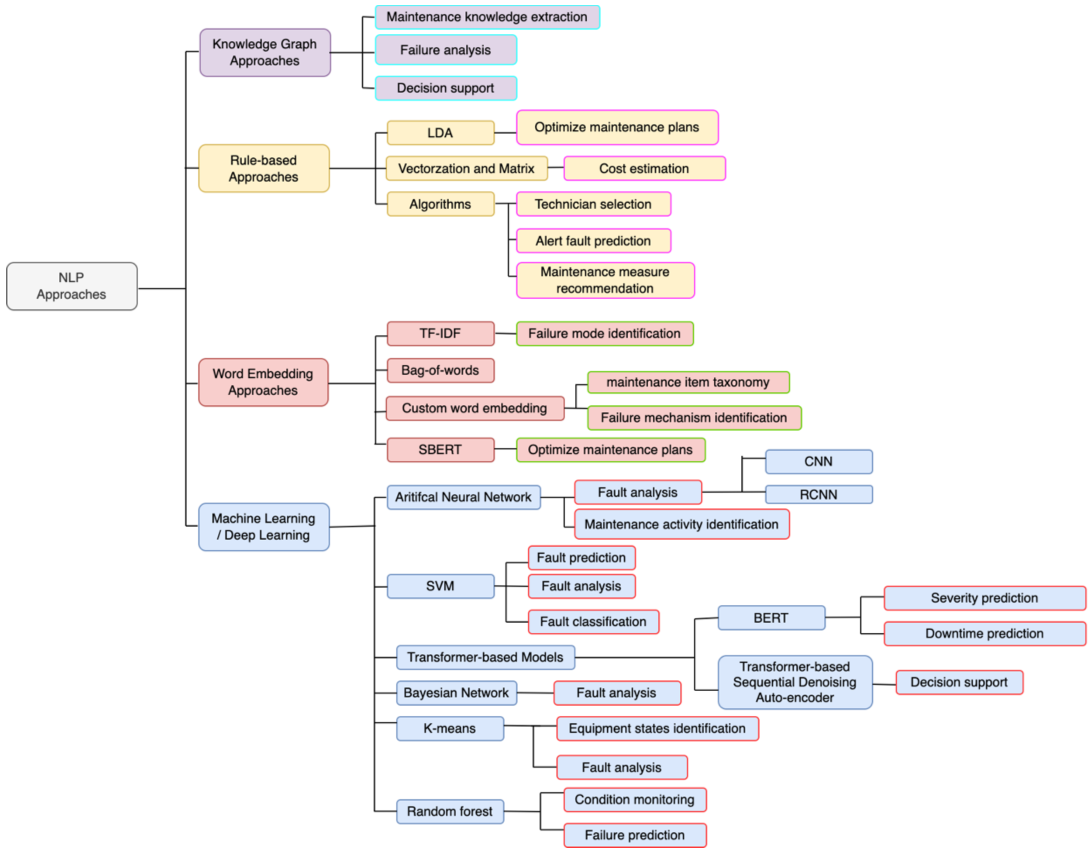

Fig. 4. Classifications and applications of NLP approaches in industrial maintenance.

Table 1. Summary of NLP approaches in selected publications.

Study	Topic	Categories	Data size	Data source	Specific purpose	Evaluation measures
Ansari (2020)	Maintenance cost estimation	Predictive maintenance	27000 entries of reports	A manufacturer company	Equipment components of 21 production line	
Ansari et al. (2021)	Downtime prediction, selection of the best-fit maintenance technician	Predictive maintenance	Digital shift book from 3 years	Automotive industry	CNC machines	Mean failure detection time Median absolute error (MAE) for downtime prediction
Nota et al. (2022)	Alert fault prediction	Predictive maintenance	30000 maintenance logs	A manufacturer in Italy	Unknown	
Sala et al. (2022)	Optimize maintenance plans	Predictive maintenance	340 interventions from maintenance reports	A manufacturer in Italy producing bottling and package machines	Unknown	The coherence and perplexity indexes
Yuan et al. (2020)	Maintenance measure recommendation	Predictive maintenance	10461 defect data	Unknown	Unknown	
Ozturk et al. (2022)	Optimize maintenance plans	Predictive maintenance	2408 entries of descriptions	Industry partners	Unknown	
Bhardwaj et al. (2022)	Failure mechanism identification, maintenance item taxonomy	Predictive maintenance	11682 maintenance logs	Eight systems (such as high-pressure mud system (mud-pump), drill rig hosting system, etc.,) across 31 oil rigs	Unknown	Adjusted ranked index (ARI) and the silhouettes score (SSc)

Table 1. (Continued)

Study	Topic	Categories	Data size	Data source	Specific purpose	Evaluation measures
Pereira (2020)	Failure mode identification	Predictive maintenance	Unknown	Unknown	Unknown	
Xu (2020)	Fault prediction	Predictive maintenance	2000 entries involving 6 failures	Unknown	Unknown	Accuracy
Gunda et al. (2020)	Fault analysis	Predictive maintenance	55000 maintenance records	Collected from 880 sites owned-operated by 6 industry partners across the U.S.	PV inverters	
Rahman (2021)	Failure classification	Predictive maintenance	7668 maintenance records of a tube filling and packaging machine	Collected from a leading personal care product manufacturer in Bangladesh	Unknown	Precision, recall, F1-score
Yang (2020)	The identification of equipment degradation states	Predictive maintenance	628 maintenance records	Collected from 26 excavators used for mining ores in 6 sites	Buckets	
Li et al. (2020)	Fault analysis	Predictive maintenance	Unknown	Collected from State Grid Corporation of China	Unknown	
Chang et al. (2018)	Fault analysis	Predictive maintenance	Unknown	Chinese aircraft maintenance data	Unknown	Accuracy
Ademujimi (2021)	Fault analysis	Predictive maintenance	770 entries of maintenance logs	Unknown	Power supply (UPS) units	
Navinchandran et al. (2021)	Identificatio of Critical maintenance activities	Predictive maintenance	47797 maintenance work orders (MWOs) of dataset A, 13268 MWOs OF dataset B, and 3427 MWOs dataset C	Collected from Automotive, lighting and automotive supplier industry	Unknown	Recall
Zhang et al. (2020)	Fault analysis	Predictive maintenance	1860 defect records	Unknown	Unknown	Accuracy
Xu et al. (2021)	Fault analysis	Predictive maintenance	More than 50000 failure texts	A civil aircraft	Unknown	Accuracy, precision, recall, F1-score
Naqvi et al. (2022)	Decision support	Corrective maintenance	5485 maintenance work orders	From maintenance operations on mining excavators	Unknown	
Usuga Cadavid et al. (2022)	Severity prediction and downtime prediction	Predictive maintenance	22709 reports of Company A, 9993 reports of Company B, 5317 reports of Company C	Aeronautics (dataset A and C), Electronics (dataset B)	Unknown	MCC (Matthew's correlation coefficient)
Usuga Cadavid et al. (2020)	Severity prediction and downtime prediction	Predictive maintenance	20000 maintenance logs	From one manufacturing company	Unknown	Accuracy, sensitivity and MCC (Matthew's correlation coefficient)
Blanco-M et al. (2019)	Condition monitoring	Predictive maintenance	Unknown	Wind turbines	Generator and gearbox	Accuracy and F-score
Dangut (2021)	Failure prediction	Predictive maintenance	3727 failure texts	Aircraft central maintenance system	Unknown	Precision, recall, F1-score, and ROC curves
Zhou et al. (2021)	Fault analysis	Predictive maintenance	Unknown	Aircraft	Unknown	Accuracy, precision, recall, F1-score
Jin et al. (2021)	Maintenance knowledge extraction	Predictive maintenance	Unknown	Unknown	Diesel engine	Entity extraction: Accuracy, precision, recall, F1-score Relation extraction: F1-score
Bian et al. (2021)	Maintenance knowledge extraction and fault analysis	Predictive maintenance	Unknown	Collected from an excavator	Unknown	Accuracy and F1-score
Chaterjee (2021)	Decision support	Predictive maintenance	A specific domain knowledge graph containing 537 nodes and 1059 relations	Operational wind farm	Wind turbines	Accuracy

3.1. Rule-based approaches

Approaches that rely on pre-determined rules and execute operations based on these rules are commonly referred to as rule-based approaches. In the field of natural language processing (NLP), rule-based approaches often make use of linguistic and statistical concepts to derive various features. Among the 27 papers analyzed in this study, four of them utilized rule-based approaches.

Ansari [19] introduced a text understanding framework known as TextPlan, designed for the estimation and analysis of maintenance costs. The framework involved vectorizing relevant elements from maintenance reports into numeric data, enabling the interpretation of costs, sentiments, and textual associations. This method is able to extract meaningful information and derive useful knowledge to maintenance planners, but one limitation is that the sentiment opinion label is required to evaluated by domain experts, which may be time-consuming. Another study by Ansari's research team explored the application of a similarity algorithm for selecting the most suitable maintenance technician [20]. When a maintenance issue arose, the similarity algorithm calculated the resemblance between the problem and the areas of expertise of the technicians, ultimately selecting the technician with the most relevant experience for the task. The use of algorithms is a common practice in rule-based approaches. This paper also addressed downtime prediction task with the use of word embedding approaches like Bag of Words (BOW) and Term Frequency-Inverse Document Frequency (TF-IDF). Some other approaches such as Random Forest (RF), k-Nearest-Neighbour (kNN), Support Vector Machine (SVM), and a Deep Neural Network (DNN) were selected as prediction models. Additionally, this paper addressed the task of downtime prediction using word embedding approaches such as Bag of Words (BOW) and Term Frequency-Inverse Document Frequency (TF-IDF). Several other approaches, including Random Forest (RF), k-Nearest-Neighbor (kNN), Support Vector Machine (SVM), and a Deep Neural Network (DNN), were selected as prediction models. By analyzing the data from digital shift books in the automotive industry, the Overall Equipment Efficiency (OEE) was enhanced by over 5%, indicating the effectiveness of the proposed method.

Nota et al. [21] developed a system based on algorithms that enabled technicians to input maintenance records and receive an alert level indicating the operational status of industrial systems. This information was valuable for technicians in scheduling appropriate maintenance activities. The algorithm used in this system is straightforward and reads the input text character by character, making it suitable for simple maintenance tasks. Sala et al. [22], on the other hand, employed topic modelling, specifically the Latent Dirichlet Allocation (LDA) algorithm, to analyze maintenance texts. Topic modelling is a widely used technique for extracting specific topics from textual data. The use of topic modelling can assist technicians in enhancing maintenance plans by extracting crucial information from maintenance texts. Furthermore, Sala et al. [22] pointed out that the LDA algorithm used in their study contributed to improved transparency and interpretability of maintenance analysis. Yuan et al. [24] proposed the use of the Simhash algorithm [25], which mapped the original maintenance text data into 64-bit binary fingerprints. Subsequently, they calculated similarity using the Hamming distance to identify relevant failure solutions. These papers demonstrate that rule-based approaches can effectively employ algorithms to accomplish simple maintenance tasks.

In summary, rule-based approaches in NLP provide a high level of explanation and transparency. They can effectively address simple maintenance tasks by utilizing a few algorithms. However, these approaches may require skilled experts to define the system rules in advance, which can be time-consuming and costly. It is important to consider the expertise and resources required when employing rule-based approaches in industrial maintenance. Future research could explore the development of automated methods for rule generation to alleviate the burden of manual rule definition and increase the efficiency of rule-based approaches in maintenance applications.

3.2. Word embedding approaches

Word embedding approaches are used to give a word representation for text analysis. Common word embedding models include TF-IDF [26], bag-of-words [27], Word2Vec [28], FastText [29], and others. Several papers have applied word embedding techniques in the initial stages of maintenance tasks to capture semantic information. However, three publications primarily used word embedding approaches for text clustering tasks related to predictive maintenance. Pereira [30] earlier examined TF-IDF to obtain relative frequency for failure mode identification, which proved that it was an effective method. Ozturk et al. [31] employed SBERT sentence embeddings [32] to learn meaningful semantics and used cosine similarity to compare with a threshold for creating maintenance event clusters, with the aim of optimizing maintenance plans. Another paper by Bhardwaj et al. [33] proposed a custom word embedding model for clustering equipment taxonomy and identifying failure mechanisms. They first used a skip-gram model to incorporate contextual information and then designed two clustering measures, within-class dissimilarity and between-class similarity, to optimize clustering based on two sources of information. Both methods are unsupervised, and they yield high-quality embedding vectors without requiring a large number of manual annotations on the datasets. The second method incorporates external information to optimize the model, potentially improving accuracy compared to the first one. In summary, word embedding approaches have demonstrated the ability to effectively capture semantic information, even in scenarios with limited data samples. However, it's important to note that these approaches are typically suited for specific tasks like failure identification. For future research, exploring methods to learn richer semantic representations and achieve more precise clustering in maintenance records represents a promising avenue.

3.3. Machine learning/deep learning approaches

Machine learning and deep learning have been widely used in processing maintenance texts in recent years. They leverage data, learn from data and make predictions or make decisions from data. 16 publications used the following methods in machine learning/deep learning:

- **Support vector machine (SVM):**

Support Vector Machine (SVM) is a widely used supervised machine learning method for classification and regression tasks, known for its high accuracy [34], [35], [2]. In the context of industrial maintenance, researchers have explored the application of SVM in various scenarios.

Xu et al. [36] utilized a multi-class SVM based on a binary tree approach to predict failure categories. Their study demonstrated the effectiveness of SVM in categorizing maintenance records into different failure types. Gunda et al. [37] focused on extracting topics related to common failure modes in photovoltaic (PV) inverters using a machine learning-based approach. They employed SVM to classify maintenance records specific to inverters and subsequently employed Latent Dirichlet Allocation (LDA) to extract topics associated with failures. The combination between supervised and unsupervised approach achieved good results. Rahman [38] also worked on identifying failure indications in maintenance records, employing SVM to determine whether a record indicates a failure event.

In contrast, Ansari et al. [20] investigated the task of downtime prediction and compared the performance of SVM with other methods. They found that SVM did not achieve optimal results for this specific prediction task. While SVM generally offers good accuracy and efficient predictions for binary classification problems, it may encounter challenges when dealing with multi-class classification problems commonly encountered in practical maintenance scenarios.

It is worth noting that SVM has demonstrated successful applications in datasets with limited data points. It is particularly effective when the dataset is small. However, in many real-world maintenance problems, multi-class classification is required, and SVM may not yield satisfactory results on its own.

To overcome the limitations of SVM in multi-class classification tasks, researchers have explored alternative approaches such as ensemble methods, deep learning, or other supervised learning algorithms specifically designed for multi-class problems. These approaches offer potential avenues for improving the performance of maintenance classification tasks involving multiple failure categories.

In summary, SVM stands as a powerful tool with high accuracy and fast prediction capabilities for addressing certain binary classification problems in maintenance. However, when confronted with multi-class classification problems, alternative methods should be considered to achieve optimal results. Further research can explore the development of hybrid models or the integration of SVM with other techniques to enhance the performance of multi-class maintenance classification tasks.

- **K-means:**

K-means is a widely used clustering algorithm within unsupervised learning techniques. It operates by grouping data points based on their proximity, typically measured using the Euclidean distance metric [39]. In the context of maintenance record analysis, Yang et al. [40] introduced a Convolutional Neural Network-based (CNN-based) approach for clustering maintenance records. CNNs were employed for feature extraction, and K-means clustering analysis was subsequently applied to cluster the text data. This approach was utilized to identify equipment degradation states. Likewise, Li et al. [41] also employed K-means clustering analysis for fault categorization in maintenance records. The K-means algorithm is favored for its simplicity in principle, high explanatory power, and ability to process large volumes of data efficiently.

However, it is important to note that K-means clustering has certain limitations. It assumes that the clusters are spherical and have equal variance, which may not hold in all scenarios. Additionally, K-means is sensitive to the initial selection of centroids and can converge to local optima. Therefore, researchers often explore enhancements such as variations of K-means or alternative clustering algorithms to overcome these limitations and improve clustering performance.

In summary, K-means clustering is a popular and effective algorithm for unsupervised clustering tasks in maintenance analysis. Its simplicity and interpretability make it a suitable choice for processing large datasets. Researchers continue to explore variations and improvements to the K-means algorithm to address its limitations and enhance its performance in various maintenance clustering applications.

- **Bayesian network:**

A Bayesian network is probabilistic graphical model that give knowledge representations and relationships with conditional probabilities via a directed acyclic graph [42]. It can be utilized for knowledge inference, classification, prediction, etc. Chang et al. [43] trained the Bayesian failure network to predict failures after using text clustering and failure occurrence sequence mining to classify the failures into six types. Ademujimi [44] introduced Bayesian Network to improve the accuracy of fault analysis and performed fusion-learning which combined sensor numerical data and maintenance reports. Naive Bayes, as a special case of Bayesian network which assumes that the features are conditional independent, was leveraged in another line of work [38] for classifying whether the maintenance records show a failure or not. Bayesian network can predict failure categories based on semantic and logical understanding, but it is difficult to identify the novel failure categories.

In summary, Bayesian networks offer a powerful framework for knowledge representation and probabilistic reasoning in maintenance analysis. Their ability to capture complex relationships and make predictions based on conditional probabilities makes them valuable for various maintenance tasks. Researchers continue to explore and refine Bayesian network-based approaches to enhance their performance, address limitations, and tackle emerging challenges in the field of maintenance analysis.

- **Artificial neural network (ANN):**

In recent years, artificial neural networks (ANNs) have gained immense popularity across various disciplines. ANNs are mathematical models designed to mimic the structure and function of the human nervous system, and they are used for estimating or approximating complex functions. ANNs consist of interconnected neurons with associated weights, forming a hierarchical structure. ANNs have been harnessed for natural language processing (NLP) tasks [45] with significant potential, making them valuable in the context of industrial maintenance. From the selected papers, Navinchandran et al. [46] concentrated on identifying significant relationships with selected Key Performance Indicators (KPIs), such as time or cost. They compared the performance of explanatory models, including Decision Trees (DT) and Neural Networks. The results indicated that DT achieved better performance regardless of the number of target classes. Zhang et al. [47] proposed a text classification model that trained Convolutional Neural Networks (CNN) and Recurrent Convolutional Neural Networks (RCNN) with Bayesian optimization to classify transformer defects. Xu et al. [48] conducted research using CNNs and embedded prior knowledge into the network based on Cloud Similarity Measurement to improve prediction results. While the selected papers primarily focused on feedforward neural networks, it's important to note that feedback neural networks, such as Recurrent Neural Networks (RNNs), have not been extensively explored for their effectiveness in analyzing maintenance logs. RNNs, especially variants like Long Short-Term Memory (LSTM) [50] and Gated Recurrent Unit (GRU) [51], have demonstrated strong performance in modeling sequences of texts [49]. These architectures have excelled in various tasks [52] and could offer valuable insights and performance enhancements when applied to the analysis of maintenance logs.

In summary, artificial neural networks (ANNs) have become popular due to their capability to model complex relationships and approximate functions. In the realm of industrial maintenance, ANNs, including CNNs, RCNNs, and RNNs, have demonstrated promise across various tasks, encompassing classification, prediction, and sequence modeling. However, there remains a need for further exploration and investigation to fully harness the potential of feedback neural networks like RNNs in the analysis of maintenance logs. Ongoing research and experimentation, exploring different ANN architectures and their combinations, will play a pivotal role in advancing the capabilities of ANNs within the field of industrial maintenance.

- **Transformer-based models:**

Transformers represent a revolutionary architectural innovation for processing sequence-to-sequence texts, and they have largely surpassed RNNs in performance in many cases. Transformers rely on a self-attention mechanism that enables them to focus on relevant features within texts and generate meaningful semantic representations. Among the selected papers, Naqvi et al. [53] utilized a pretrained Transformer-based Sequential Denoising Auto-Encoder (TSDAE) for solving maintenance problems. TSDAE is a model that has achieved state-of-the-art performance compared to other approaches like masked language models. Cadavid et al. [54], [55] employed the BERT model [14] in the French language to predict the severity of failures, the causes of breakdowns, and the time of breakdowns. Transformers have consistently demonstrated superior performance in processing sequence-to-sequence texts compared to RNNs, owing to their self-attention mechanism and capacity to capture relevant features and semantic representations effectively. Moreover, transformer-based models excel at handling large-scale datasets. Further research and exploration of transformer-based architectures and their application in the field of industrial maintenance will contribute to advancing the state-of-the-art in text processing and analysis.

- **Random forest (RF):**

Random Forest (RF) is an ensemble learning method that employs multiple randomized decision trees, making it particularly well-suited for handling large-scale datasets. By combining the predictions of individual decision trees, RF can provide accurate and robust results. However, the interpretability of RF models can be challenging due to the complex interactions between these decision trees. In the context of industrial maintenance, Blanco-M et al. [56] utilized RF for the classification of failures about wind turbines aiming to conduct condition monitoring. Their work indicated that RF can help identify which words are related to specific failures, shedding light on the factors contributing to those failures. In 2021, Ansari et al. [20] conducted research on downtime prediction and evaluated various models, including RF. Their study aimed to explore the effectiveness of different approaches in predicting downtime and assess the performance of RF in comparison to other models. Dangut et al. [57] ensembled a RF model after vectorizing the texts and identifying the patterns using TF-IDF and Word2Vec in order to predict aircraft component failure.

RF models have gained popularity in maintenance research due to their ability to handle diverse data types and capture complex relationships within the data. The ensemble nature of RF helps mitigate overfitting and provides robust predictions. However, interpreting the specific factors contributing to the predictions made by RF can be challenging, as it involves analyzing the collective behavior of multiple decision trees. While RF models have been applied successfully in various maintenance tasks, it is essential to consider the trade-off between predictive performance and interpretability. Researchers and practitioners should carefully evaluate the requirements of their specific maintenance problem and select the appropriate modeling approach accordingly.

In summary, RF is an ensemble learning method that utilizes randomized decision trees and is well-suited for handling large-scale data sets in industrial maintenance. The study by Ansari et al. [20] provides insights into the application of RF in the context of downtime prediction. As RF models offer high predictive accuracy, they can be valuable tools in maintenance analytics. However, their interpretability should be considered when analyzing and understanding the factors driving the predictions. Further research can focus on developing techniques to enhance the interpretability of RF models without sacrificing their predictive power.

- **Stacking ensemble methods:**

Stacking ensemble methods, as demonstrated by Zhou et al. [58], offer a powerful approach to improving the accuracy of failure diagnosis in the field of industrial maintenance. This method leverages the strengths of multiple base models and combines them into a more robust predictive model. Two-layer stacking ensemble model was used with KNN, SVM, gradient boosting decision tree and naïve Bayes as the basic learner and random forest as the top layer. The use of stacking ensemble methods in failure diagnosis has several advantages. Firstly, it leverages the diversity of models, which can capture different aspects of the failure patterns and contribute complementary information. Secondly, the stacking ensemble approach can effectively handle complex relationships and dependencies within the data, enabling more accurate diagnosis. Finally, the ensemble method provides a framework for model selection and combination, allowing researchers to leverage the strengths of various models and optimize the overall performance.

While the stacking ensemble method proposed by Zhou et al. [58] showed promising results in failure diagnosis, further research is needed to explore its application in different maintenance scenarios and evaluate its performance on diverse datasets. Additionally, investigations into the interpretability of stacking ensemble models can provide insights into the reasoning process behind the failure diagnosis and enhance the transparency of the decision-making process.

In conclusion, stacking ensemble methods have demonstrated their effectiveness in improving the accuracy of failure diagnosis. The study by Zhou et al. [58] showcased the application of a two-layer stacking ensemble model, combining diverse base learners and a random forest top layer. This approach allowed for enhanced predictions by leveraging the strengths of individual models and addressing their limitations. Future research can explore the adaptability of stacking ensemble methods to various maintenance problems and investigate techniques to interpret and explain the decision-making process of these models.

3.4. Knowledge graph approaches

The effective maintenance of industrial systems requires the accumulation of considerable expert knowledge, often stored in maintenance reports. This knowledge can be extracted and used to construct knowledge graphs (KGs). Recently, KGs have garnered significant attention as powerful tools in natural language processing. KGs can be viewed as semantic networks that encompass entities and their relationships, all expressed as textual data. The application of KGs can facilitate effective decision-making and, at times, lead to savings in maintenance time and costs. From the selected literature, 3 papers are related with knowledge graphs from which 2 papers focus on the construction of knowledge graphs and 1 paper involves the application of KGs. Jin et al. [59] extracted entities and bidirectional relations to construct diesel engine maintenance knowledge graph. Pre-trained language model BERT was applied to obtain the feature representation (i.e. named entity recognition) and Bi-LSTM-CRF framework [60] was used to extract the entities and relations. Bian et al. [61] used a rule-based named entity recognition method for extraction of entities and relations, and then trained CNN in order to classify the entities for the final construction. As for the application of KGs, Chatterjee [62] proposed an intelligent question-answering system based on the reasoning of knowledge graph database, where a maintenance problem is input by a technician and all the possible maintenance actions will be retrieved as answers. Reasoning methods including Seq2Seq models and transformer models were used. In general, the construction of maintenance KGs relies on automated methods, predominantly employing deep learning techniques. However, these methods often lack transparency and interpretability, posing a challenge in ensuring the accuracy of the extracted knowledge. Additionally, considering the graph structure information alongside textual information (entities and relations) in KG reasoning remains an unexplored area.

KGs offer a valuable approach for transforming textual information into actionable insights, making them a valuable tool for knowledge accumulation in industrial maintenance. KGs can empower technicians and professionals to conduct maintenance tasks more effectively by providing structured access to critical information, regardless of the data volume. Future research can explore KG construction with better transparency and interpretability and enhance reasoning capabilities through graph structure.

4. Data and evaluation measures

This section describes datasets these publications employed and summarizes evaluation measures they took, which gives an overview for researchers to consider how NLP approaches are used based on datasets and help provide potential solutions for manufacturing industries.

4.1. Datasets

Table 2 offers statistics related to the data size in these studies. One observation is that almost all the datasets come from maintenance reports/logs and maintenance work orders (MWOs) except for 7 unknown datasets. Moreover, it revealed that most publications have access to large datasets containing more than 1000 entries, with ten datasets even surpassing 10,000 entries. There is only one instance where a paper relies on a dataset containing less than 100 entries. In fact, many companies possess small and incomplete datasets, making it imperative to explore how to process and extract insights from such small-scale datasets effectively. This issue presents a valuable area for exploration and research in the field of industrial maintenance.

Table 2. Number of studies by different data sizes.

Data size	Number of studies
0 - 200	0
200 - 500	1
501 - 1000	2
1001 - 2000	2
2001 - 5000	3
5001 - 10000	4
10000+	10

4.2. Evaluation measures

It is important to highlight that out of the 27 publications reviewed, 10 of them did not include any explicit evaluation measures. These papers primarily focused on demonstrating how to perform maintenance tasks without providing details on datasets or evaluation metrics. Unfortunately, such papers do not offer much value in terms of providing references for future research endeavours.

For literature that does include evaluation measures, it is essential to establish standardized approaches for evaluating different tasks. By adopting consistent evaluation measures, researchers can ensure comparability and facilitate meaningful comparisons across different studies. Standardization can also enhance the reliability and reproducibility of research outcomes in the field of maintenance.

To establish standardized evaluation measures, the following strategies can be considered:

1. **Define common evaluation metrics:** Researchers should agree upon a set of commonly used evaluation metrics that are appropriate for specific maintenance tasks. For instance, in the context of fault diagnosis, metrics such as accuracy, precision, recall, and F1 score can be employed. By adopting consistent metrics, researchers can evaluate the performance of different methods and facilitate comparisons.

   - *Failure/cause prediction:*

     Precision, recall and F1 score will be utilized as the evaluation metrics. Precision talks about how precise/accurate the model is out of those predicted positive, how many of them are actual positive (1):

     $$Precision = \frac{TruePositive}{TruePositive+FalsePositive} \quad (1)$$

     Recall calculates how many of the Actual Positives our model capture through labelling it as Positive (True Positive) (2):

     $$Recall = \frac{TruePositive}{TruePositive+FalseNegative} \quad (2)$$

     F1 score is a function of Precision and Recall (3):

     $$F1 = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (3)$$

- *Text clustering and topic extraction:*

  There are three evaluation measures for the tasks, i.e., the coherence and perplexity indexes that iare used for evaluating topic models, *the adjusted ranked index (ARI)* and *the silhouettes score (SSC)* that are used for clustering.

2. **Establish benchmark datasets:** It is crucial to curate and share benchmark datasets that represent real-world maintenance scenarios. These datasets should encompass diverse examples and cover a wide range of challenges and complexities encountered in industrial maintenance. By using standardized datasets, researchers can evaluate their methods on a common ground, enabling fair comparisons and promoting advancements in the field.
3. **Encourage replication studies:** Replication studies play a vital role in verifying and validating research findings. Researchers should be encouraged to replicate existing studies using standardized evaluation measures and datasets. Replication studies can help confirm the effectiveness and generalizability of proposed approaches and contribute to the establishment of best practices in maintenance research.
4. **Promote transparency and reporting:** Researchers should provide clear and detailed descriptions of their experimental setups, including the dataset used, pre-processing steps, model configurations, and evaluation procedures. Transparent reporting allows other researchers to reproduce the results and ensures transparency in the research process.

By implementing these strategies, the field of maintenance research can establish standardized evaluation measures, leading to more robust and reliable findings. Standardization facilitates cumulative knowledge building, promotes advancements, and guides future research directions.

It is important for researchers to recognize the significance of including evaluation measures in their publications. By doing so, they contribute to the collective knowledge in the field and provide valuable references for future research endeavours.

5. Conclusion

This paper presents a systematic literature review that focuses on the main applications of natural language processing (NLP) approaches in maintenance. The review includes the categorization and applications of NLP approaches and provides an analysis of their merits and limitations. Four research questions were formulated to guide the review process, ensuring comprehensive coverage of relevant studies.

The findings reveal that maintenance measures can pertain to the entire production line or specific equipment. In terms of datasets, it was observed that large datasets were predominantly utilized, while exploration of small datasets was limited. It is worth noting that some papers lacked sufficient details regarding their datasets and evaluation measures, rendering them less instructive and difficult to learn from. Additionally, the diversity of evaluation measures used across different papers hampers direct comparisons between NLP approaches.

Moreover, this systematic literature review highlights the significance of predictive maintenance as a promising strategy that has garnered considerable attention from researchers. By avoiding unnecessary maintenance, predictive maintenance offers substantial cost reductions and significant improvements in production efficiency. NLP approaches have played a crucial role in enhancing failure diagnosis accuracy, decision-making capabilities, and reducing maintenance time.

Despite the progress made in applying NLP methods to industrial maintenance, there are still notable research gaps to address. Several potential future research directions can be considered, including:

1. **Investigation of small-scale textual datasets:** While many studies have focused on large datasets, it is important to acknowledge that numerous manufacturing companies possess small and incomplete datasets. Future research should explore the challenges and opportunities associated with small-scale data, developing tailored NLP approaches for such scenarios.

2. **Comparison of different machine/deep learning approaches:** Further studies should compare the performance of various machine learning and deep learning algorithms using datasets of different sizes. Such comparisons can provide insights into the strengths and weaknesses of different approaches and guide the selection of suitable models for specific maintenance tasks.
3. **Development of novel algorithms and interpretable models:** Researchers should explore the development of innovative machine learning and deep learning algorithms that are specifically tailored for maintenance applications. Additionally, there is a need for interpretable machine learning and deep learning models that can provide transparent explanations for the decisions made, aiding in the acceptance and trustworthiness of the generated results.
4. **Precise and effective evaluation in real industrial environments:** Future research should prioritize the evaluation of NLP approaches in real-world industrial settings. This entails conducting experiments and validations in actual maintenance scenarios, considering the practical constraints and challenges faced in industrial environments. Such evaluations will provide more accurate and reliable assessments of the performance and applicability of NLP approaches.

By addressing these research gaps and pursuing the suggested future directions, the field of NLP in maintenance can advance significantly, offering tailored solutions that cater to diverse data scenarios, facilitating comparative analyses, promoting interpretability, and ensuring the practical relevance and effectiveness of NLP approaches in real industrial environments.

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